

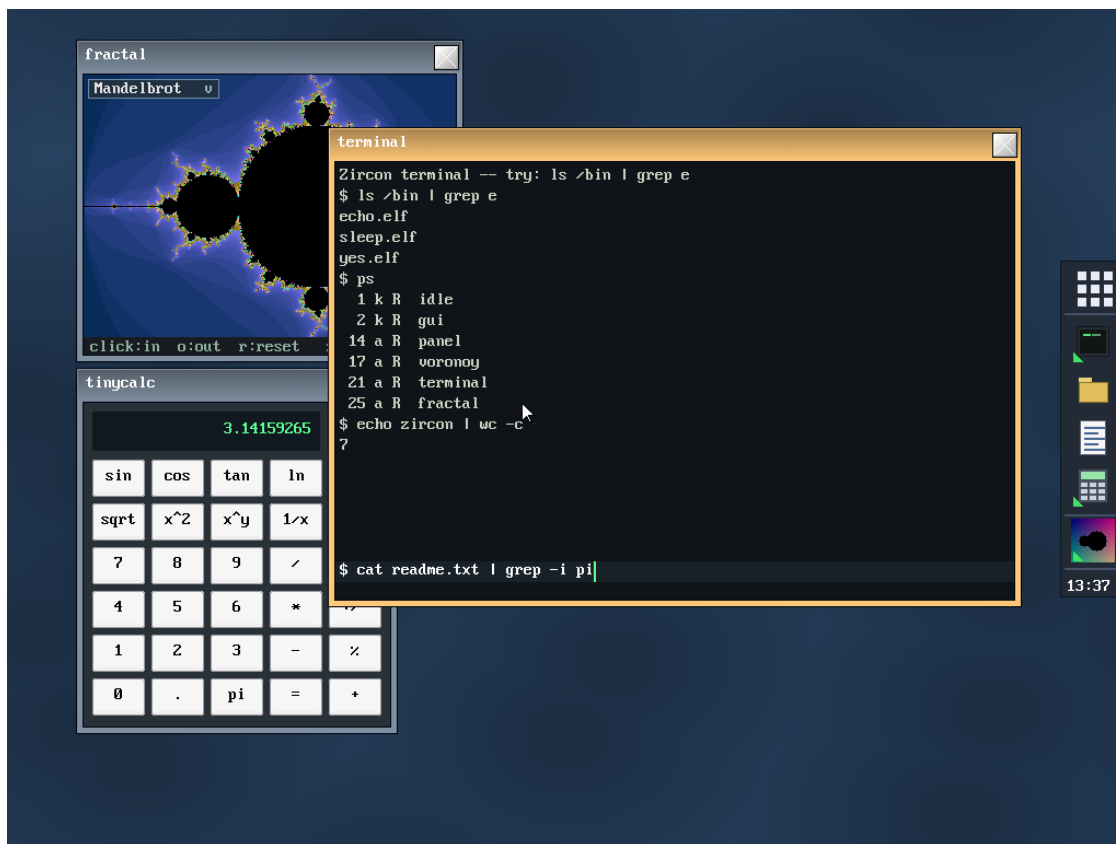
# Onyx — Project Overview

*Onyx · multi-process OS on Raspberry Pi 4*

## 1. In one sentence

Onyx is a **small multi-process operating system** for the **Raspberry Pi 4** (AArch64), built **on top of Circle** ([rsta2/circle](#)), which serves as its HAL and driver stack. It loads **ELF programs from the SD card** and runs them as **applications isolated from one another**, each in its own page table. The whole thing is driven by the **Onyx desktop** made up of a software compositor, a taskbar, a launcher, a terminal, a file manager, and about thirty applications.

It **runs on real Raspberry Pi 4 hardware** (not just in emulation).



*The Onyx*

*desktop: panel/launcher on the edge, composited windows (fractal browser, active terminal, calculator), Voronoi wallpaper. (Simulated screenshot, rendered from the real skins/font/icons.)*

## 2. Philosophy: "own the top, reuse the bottom"

Circle is a *single-process, single-address-space, cooperative* bare-metal framework ("your application **is** the system"). Onyx **keeps the bottom half** of Circle (which is excellent) and **replaces its "operating system" half**:

| Layer                             | Decision           | Detail                                   |
|-----------------------------------|--------------------|------------------------------------------|
| Reset → EL1, MMU activation, GIC, | <b>Keep Circle</b> | Our entry point is <code>main()</code> ; |

| Layer                                                                         | Decision                                        | Detail                                         |
|-------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------|
| C++ constructors                                                              |                                                 | everything below it is already done.           |
| 64 KB page allocator ( <code>palloc/pfree</code> )                            | <b>Keep</b> — it's our frame allocator          | <code>CMemorySystem::Get()</code> .            |
| GIC-400, timer, mailbox, EMMC+FatFs, USB, Ethernet, framebuffer               | <b>Reuse</b>                                    | This is the whole point of building on Circle. |
| <code>CScheduler/CTask</code> scheduler                                       | <b>Rewrite the implementation, keep the API</b> | See <a href="#">kernel internals</a> .         |
| Per-process page tables, context switching, exception vectors, address spaces | <b>Write from scratch</b>                       | Doesn't exist in Circle.                       |

Practical consequence: we **link** the kernel against Circle's already-compiled static libraries (`libcircle.a`, `libfs`, `libusb`, ...) rather than copying its sources.

### 3. Main features

- **Per-process isolation via the MMU.** Each application has its own L2/L3 page table (64 KB granule), tagged by **ASID**. One application cannot see another's memory.
- **"Option C" execution model.** Applications run in **EL1** (privileged) and call the kernel's functions directly (no `SVC` trap in the common path). The trade-off: apps are isolated **from one another**, but **not from the kernel**. See §5.
- **Stable fixed-address ABI (`kapi`).** The kernel publishes a **table of function pointers** at a fixed virtual address, mapped read-only into each application. Applications call the kernel through this table → **an application binary keeps working without recompilation** when the kernel changes (*append-only* contract, current ABI version: **21**).
- **Full graphical desktop.** 32-bit software compositor, window manager, toolkit of kernel-drawn widgets (buttons, checkboxes, sliders, text fields, scroll bars, icons...), windows with themeable decoration, wallpaper, mouse cursor.
- **Onyx shell.** Panel/launcher (panel), app list, interactive terminal with **pipes and redirection** (`|`, `>`, `>>`, `<`), file manager, and a collection of command-line tools in `/bin`.
- **Application catalog.** Text editor, spreadsheet, scientific calculator, drawing program, fractal browser, calendar, task manager, theme editor, and many games (Tetris, Snake, 2048, Minesweeper, Sokoban, Pong, Game of Life, SameGame).
- **USB input devices.** Keyboard and mouse (HID), with hot-swappable keyboard layout switching (US, UK, DE, FR, ES, IT, Dvorak).
- **Networking (WLAN).** TCP/IP stack + on-board Wi-Fi (BCM4343 / `wpa_supplicant`) brought up on the primary core, TCP sockets exposed to apps through the ABI, an **IRC client**, and NTP clock synchronisation. (A dedicated network core is a planned next step.)

### 4. The architecture at a glance

|                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Applications (ELF EL1, isolated by ASID)<br>panel · applist · terminal · filer · tinypad · games · /bin/*<br> <br>call the kernel through <code>kapi.h</code> (inline wrappers) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| ▼ kapi ABI table (at 14 GB, read-only in every app)                                                                                                                                                                                                                                                                                                                                                       | ← stable contract |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| ONYX KERNEL (EL1) <ul style="list-style-type: none"> <li>• mm/ per-process address spaces, MMU, ASID</li> <li>• sched/ cooperative scheduler (replaces Circle's)</li> <li>• arch/ VBAR_EL1 exception vectors, trap frame</li> <li>• proc/ ELF64 loader</li> <li>• sys/ kapi impl., stream/stdio, debug console</li> <li>• gui/ GImage (software renderer), compositor+WM, skins, modal dialogs</li> </ul> |                   |
| CIRCLE (HAL + drivers, reused as-is)<br>palloc · GIC · timer · EMMC+FatFs · USB HID · framebuffer 2D                                                                                                                                                                                                                                                                                                      |                   |
| Raspberry Pi 4 (BCM2711, 4× Cortex-A72, ARMv8-A)                                                                                                                                                                                                                                                                                                                                                          |                   |

## 5. The "Option C" execution model

This is the most structurally defining architectural decision. After prototyping EL0 processes + system calls (SVC), the project switched to **Option C**:

- Applications run in **EL1** (the same privilege level as the kernel), **each in its own page table** tagged by ASID.
- They **call the kernel's functions directly** via the [kapi](#) ABI table — no system trap in the normal path.
- **Isolation**: applications are isolated **from one another** (a distinct ASID/TTBR0 per app; an app cannot address another's pages). The **kernel is not protected**: an EL1 app can technically touch the kernel's memory. This is the accepted trade-off in exchange for the ergonomics of direct calls.
- The EL0/SVC machinery (vectors, system-call dispatch) **still exists but is inert**; it could be reused if one wanted true EL0 applications isolated from the kernel.

## 6. Scheduling: cooperative (on hardware)

Although the kernel was initially designed to be **preemptive** (100 Hz tick), the port to hardware showed that **preemption from the IRQ does not work** in Circle's model: threads run in **EL1t** (with `SP_EL0`), while the IRQ handler runs in **EL1h** (with `SP_EL1`). A context switch from the IRQ would swap `SP_EL1`, not the thread's stack.

→ **Scheduling is therefore cooperative**. Tasks switch when they call `yield`, `msleep`, `present`, `wait`, etc. — exactly like Circle's original scheduler. Graphical applications naturally yield on each frame (via `present/msleep`), which gives the illusion of parallelism. Details in [Kernel internals](#).

## 7. Target hardware

- **Raspberry Pi 4** (BCM2711, 4× Cortex-A72, ARMv8-A) — primary target.
- **HDMI** output (32 bpp framebuffer, 1024×768 by default, configurable).
- **USB keyboard + mouse**.
- Optional **serial console** (GPIO14/15, 115200 8N1) for the boot log and exception dumps.
- The RPi 5 is deferred (I/O behind the proprietary RP1 chip via PCIe — heavier bring-up). Multi-core (SMP) is also deferred.

## 8. Repository structure

|                 |                                                                  |
|-----------------|------------------------------------------------------------------|
| ARCHITECTURE.md | original design + build manifest (historical, partly dated)      |
| README.md       | summary (partly dated: describes the "2 demos" state)            |
| docs/           | THIS documentation (overview, internals, dev/user guides)        |
| kernel/         | the Onyx kernel (see docs/02-KERNEL-INTERNAL.md)                 |
| user/           | the userland: apps (*.c), runtime (crt0.S, user.ld), /bin tools  |
| sdcard/         | ready-to-flash files for an RPi 4 (firmware, config, apps, /bin) |
| tools/          | host scripts (BMP icon generation)                               |
| circle/         | upstream Circle clone (not committed; cloned separately)         |

**Beware of legacy docs.** ARCHITECTURE.md, README.md, kernel/README.md, and sdcard/README.md describe **older states** of the project (ELO processes, preemptive scheduling, 640×480, "two demos"). Where they contradict this documentation, **these documents here (docs/)** are **authoritative** for the current state. ARCHITECTURE.md §11–§12 nevertheless remains the best reference for *why* Option C and cooperative scheduling were chosen.

## 9. Further reading

| Document                     | For whom                                  | Contents                                                               |
|------------------------------|-------------------------------------------|------------------------------------------------------------------------|
| <b>02 — Kernel Internals</b> | anyone who wants to understand the kernel | boot, memory/MMU, scheduling, exceptions, ABI, GUI, streams            |
| <b>03 — Developer Guide</b>  | anyone who wants to build/compile/extend  | toolchain, build, app model, extending the ABI, conventions, debugging |
| <b>04 — User Guide</b>       | anyone who wants to use it                | SD card, desktop, terminal, files, applications, customization         |

## 10. Quick genesis (milestones)

1. Taking control of `main()` on top of Circle's `sysinit`; serial console.
2. Replacement scheduler ("shadow" header for `CScheduler`).
3. `VBAR_EL1` vectors, trap frame, system-call path.
4. Per-process page tables (`CAddressSpace`), TTBR0/ASID switching.
5. ELF64 loader → process.
6. Framebuffer (`C2DGraphics`) + `GImage` rendering core (ported from the author's FreeBASIC `SimpleOS`).
7. Compositor + window manager; two animated demos running simultaneously.
8. **Switch to Option C** (EL1 apps + direct call) then **fixed-table ABI**.
9. Onyx desktop (panel + `applist`), stream/stdio subsystem, terminal + `/bin`, file manager.
10. Modal dialogs, themes, app-drawn wallpaper, PID management, keyboard layouts, theme editor (ABI v16).
11. Networking: WLAN bring-up + TCP/IP on the primary core, TCP socket calls (ABI v21), an IRC client, a `/bin/net` tool, and NTP clock sync.